The body's largest stores of glycogen are found in the liver and skeletal muscles. Individual glucosyl residues (ie, glucose monomers) in muscle and liver glycogen particles are linked by α(1→4) and α(1→6) glycosidic bonds and are released in the reactions of glycogenolysis. Glycogen phosphorylase catalyzes the removal of most of the glucosyl residues, although debranching enzyme also releases the single glucose molecule that remains after it repositions three residues onto an adjacent linear chain.

Binding of the hormone epinephrine to its receptor on the surface of muscle cells stimulates muscle glycogenolysis through activation of a signaling cascade (Figure 1).

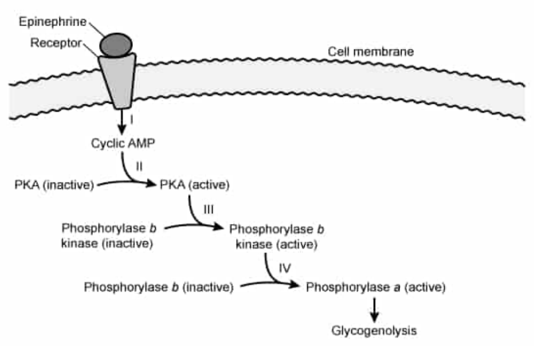
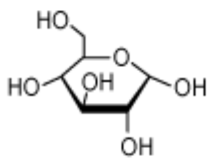


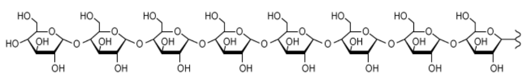
Figure 1 Stimulation of skeletal muscle glycogenolysis by epinephrine.

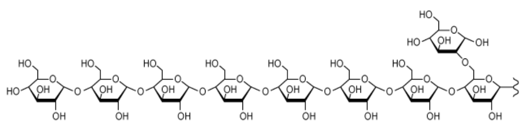
The relative signal amplification (SA) at each step (I–IV in Figure 1) can be determined by comparing the number of active signaling molecules before (M_{before}) and after (M_{after}) the step:

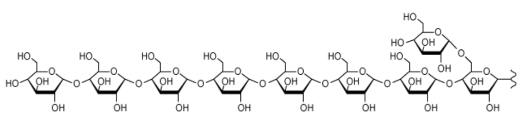
$$\text{SA} = \frac{M_{\text{after}}}{M_{\text{before}}}$$

Which structure does NOT represent a product of the reactions catalyzed by glycogen debranching enzyme?

A.  [7%]

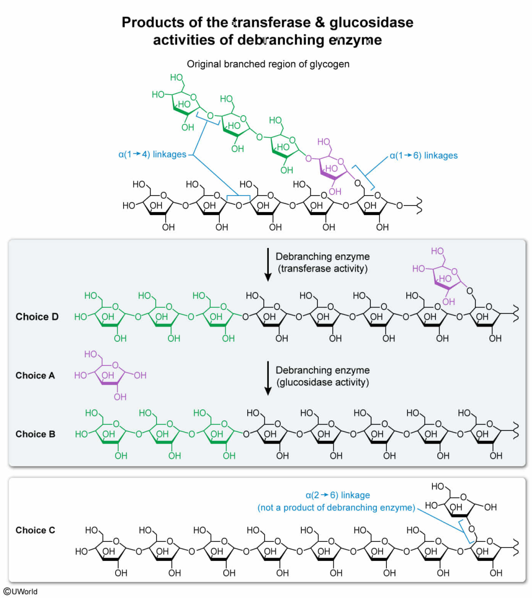
B.  [12%]

✔ C.  [60%]

D.  [19%]

Correct
60% answered correctly

Explanation:



Glycogen is **composed of** glucose monomers (**glucosyl residues**) **linked** by **α(1→4) glycosidic bonds** in **linear segments** and by **α(1→6) glycosidic bonds** at **branch points**. The enzymes glycogen phosphorylase and debranching enzyme work together to release these residues during glycogenolysis.

The question asks for the structure that does NOT represent a product of the reactions catalyzed by glycogen debranching enzyme. As indicated in the passage, glycogen debranching enzyme is a dual-function enzyme, functioning as both a transferase and a glucosidase, thereby producing three types of products:

- Single-residue glycogen branches, produced when three of four branch residues are transferred to the end of a linear segment **(Choice D)**.
- Individual glucose molecules, produced when single-residue branches are released by debranching enzyme's glucosidase activity **(Choice A)**.
- Linear glycogen segments, resulting from the combined transferase and glucosidase activities of debranching enzyme **(Choice B)**.

Therefore, the structure in Choice C, which shows a **single-residue glycogen** branch **linked via an α(2→6) glycosidic bond**, is the only structure *not* produced by the debranching enzyme.

Educational objective:

Glycogen is composed of glucosyl residues linked by α(1→4) glycosidic bonds in linear segments and α(1→6) glycosidic bonds at branch points.

Time spent:0 sec | **QID:**403532

Subject: Biochemistry | **System:** Metabolic Reactions

The body's largest stores of glycogen are found in the liver and skeletal muscles. Individual glucosyl residues (ie, glucose monomers) in muscle and liver glycogen particles are linked by α(1→4) and α(1→6) glycosidic bonds and are released in the reactions of glycogenolysis. Glycogen phosphorylase catalyzes the removal of most of the glucosyl residues, although debranching enzyme also releases the single glucose molecule that remains after it repositions three residues onto an adjacent linear chain.

Binding of the hormone epinephrine to its receptor on the surface of muscle cells stimulates muscle glycogenolysis through activation of a signaling cascade (Figure 1).

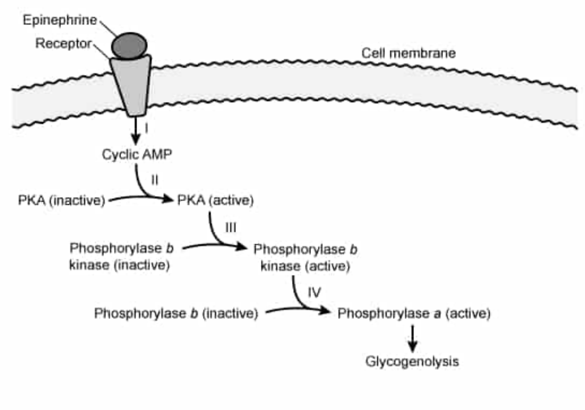


Figure 1 Stimulation of skeletal muscle glycogenolysis by epinephrine.

The relative signal amplification (SA) at each step (I–IV in Figure 1) can be determined by comparing the number of active signaling molecules before (M_{before}) and after (M_{after}) the step:

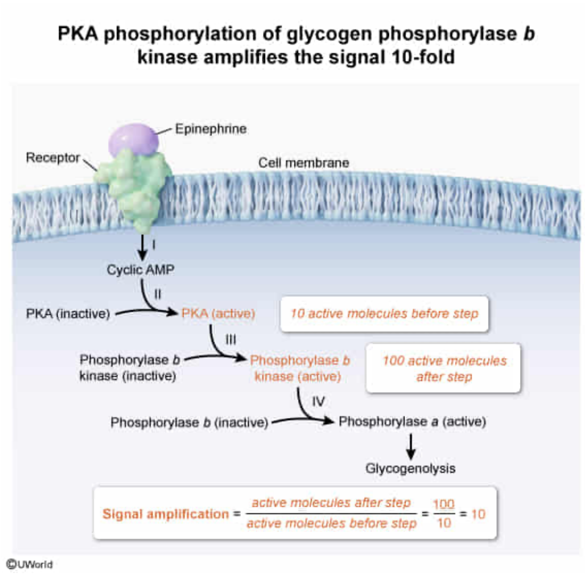
$$\text{SA} = \frac{M_{\text{after}}}{M_{\text{before}}}$$

Based on Figure 1, if a single epinephrine molecule binding to its receptor results in 10 active PKA molecules, 20 cAMP molecules, and 100 active glycogen phosphorylase *b* kinase molecules being produced, the signal amplification induced by PKA phosphorylating its target would be:

- A. $\frac{1}{5}$ [9%]
- B. $\frac{1}{2}$ [9%]
- C. 2 [14%]
- ✔ D. 10 [67%]

Correct
67% answered correctly

Explanation:



Ligand binding to a receptor protein often triggers a response via the process of signal transduction. Phosphorylation cascades are common mediators of intracellular responses to extracellular ligand binding. In these cascades, a molecule connected to the receptor activates a protein kinase. Protein kinases transfer phosphate groups from ATP molecules to proteins (ie, the enzyme phosphorylates the proteins). Through this mechanism, active protein kinases amplify the signal initiated by the ligand, resulting in a specific cellular response.

Figure 1 depicts a phosphorylation cascade initiated by epinephrine that eventually stimulates glycogenolysis, and the question asks for the signal amplification induced by PKA phosphorylating its target. Figure 1 shows that the target of PKA in this case is glycogen phosphorylase *b* kinase. The question states that an epinephrine molecule binding its receptor results in 10 active PKA molecules being produced and, subsequently, 100 active glycogen phosphorylase *b* kinase molecules. **Using** the **equation** for signal amplification (SA) provided **in the passage**, the **signal amplification produced** when PKA phosphorylates glycogen phosphorylase *b* kinase in step III **is 10**, as can be determined by comparing the number of active signaling molecules before (M_{before}) and after (M_{after}) the step:

$$\text{SA} = \frac{M_{\text{after}}}{M_{\text{before}}}$$

$$\text{SA} = \frac{100 \text{ molecules } (\text{of active glycogen phosphorylase } b \text{ kinase})}{10 \text{ molecules } (\text{of active PKA})} = 10 \text{ fold amplification}$$

(Choice A) Rather than signal amplification, this value is the ratio of cAMP to active glycogen phosphorylase *b* kinase.

(Choice B) This value is the signal amplification produced in step II of the pathway (ie, activation of 10 PKA molecules by the 20 cAMP molecules produced in step I) rather than the amplification induced by PKA phosphorylating its target in step III.

(Choice C) Instead of the signal amplification induced by PKA phosphorylating its target in step III, this value is the ratio of cAMP molecules to active PKA produced by

binding a single epinephrine molecule to its receptor.

Educational objective:
The original hormonal signal for increased glycogenolysis is amplified by the phosphorylation of enzymes in a phosphorylation cascade.

Time spent:0 sec
Subject:
Biochemistry

QID:403533
System:
Metabolic Reactions

The body's largest stores of glycogen are found in the liver and skeletal muscles. Individual glucosyl residues (ie, glucose monomers) in muscle and liver glycogen particles are linked by α(1→4) and α(1→6) glycosidic bonds and are released in the reactions of glycogenolysis. Glycogen phosphorylase catalyzes the removal of most of the glucosyl residues, although debranching enzyme also releases the single glucose molecule that remains after it repositions three residues onto an adjacent linear chain.

Binding of the hormone epinephrine to its receptor on the surface of muscle cells stimulates muscle glycogenolysis through activation of a signaling cascade (Figure 1).

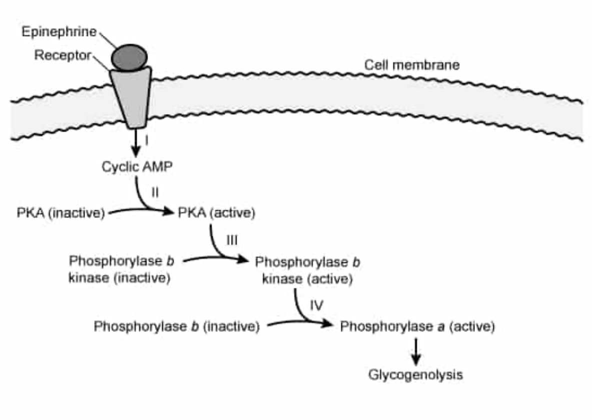


Figure 1 Stimulation of skeletal muscle glycogenolysis by epinephrine.

The relative signal amplification (SA) at each step (I–IV in Figure 1) can be determined by comparing the number of active signaling molecules before (M_{before}) and after (M_{after}) the step:

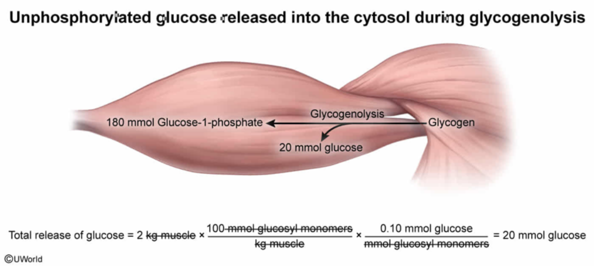
$$\text{SA} = \frac{M_{\text{after}}}{M_{\text{before}}}$$

Assuming that 2 kg of skeletal muscle are exposed to conditions that cause the cleavage of 100 mmol of glucose monomers per kilogram of muscle (some of which are phosphorylated by the cleavage event), and that 90% of these monomers are released from glycogen by glycogen phosphorylase, what is the total amount of unphosphorylated glucose released into the cytosol by this process?

- A. 10 mmol [12%]
- B. 20 mmol [60%]
- C. 90 mmol [5%]
- D. 180 mmol [21%]

Correct
60% answered correctly

Explanation:



During glycogenolysis, glucosyl monomers are released either as glucose 1-phosphate (G1P) (due to phosphorolysis by glycogen phosphorylase) or as unphosphorylated glucose (due to the glucosidase action of debranching enzyme).

The question asks for the total amount of unphosphorylated glucose released into the cytosol during glycogenolysis in 2 kg of skeletal muscle under conditions in which 100 mmol of glucosyl monomers are cleaved from glycogen per kilogram of muscle. The question states that 90% of the glucosyl monomers are released by the action of glycogen phosphorylase. These monomers are released as G1P, and the remaining 10% are released as unphosphorylated glucose by the debranching enzyme. The **total amount** of **glucosyl monomers** can be **determined by multiplying** the **release per kilogram by** the **mass in kilograms** of skeletal muscle under consideration:

$$\text{Total release of glucosyl monomers} = \text{release per kilogram muscle} \times \text{kilograms of muscle}$$

$$\text{Total release of glucosyl monomers} = \frac{100 \text{ mmol glucosyl monomers}}{\cancel{\text{kg muscle}}} \times 2\,\cancel{\text{kg muscle}}$$

$$= 200 \text{ mmol glucosyl monomers}$$

Therefore, the total amount of unphosphorylated glucose released into the cytosol during glycogenolysis in this scenario is 20 mmol:

$$\text{Total glucose release} = \text{total glucosyl monomer release} \times 10\,\%$$

$$\text{Total glucose release} = 200 \text{ mmol glucosyl monomers released} \times 0.10 = 20 \text{ mmol of glucose}$$

(Choice A) This is the amount of glucose released in 1 kg of muscle (ie, it does not take into account the mass of muscle given in the question).

(Choices C and D) These are the amounts of *G1P* released (by the action of *glycogen phosphorylase*), either in 1 kg or 2 kg of skeletal muscle, respectively.

Educational objective:
Glycogenolysis releases glucosyl units mainly as glucose 1-phosphate via the action of glycogen phosphorylation, with the remaining glucosyl units released as free glucose due to the glucosidase activity of debranching enzyme.

Time spent:0 sec
Subject:
Biochemistry

QID:403534
System:
Metabolic Reactions

The body's largest stores of glycogen are found in the liver and skeletal muscles. Individual glucosyl residues (ie, glucose monomers) in muscle and liver glycogen particles are linked by α(1→4) and α(1→6) glycosidic bonds and are released in the reactions of glycogenolysis. Glycogen phosphorylase catalyzes the removal of most of the glucosyl residues, although debranching enzyme also releases the single glucose molecule that remains after it repositions three residues onto an adjacent linear chain.

Binding of the hormone epinephrine to its receptor on the surface of muscle cells stimulates muscle glycogenolysis through activation of a signaling cascade (Figure 1).

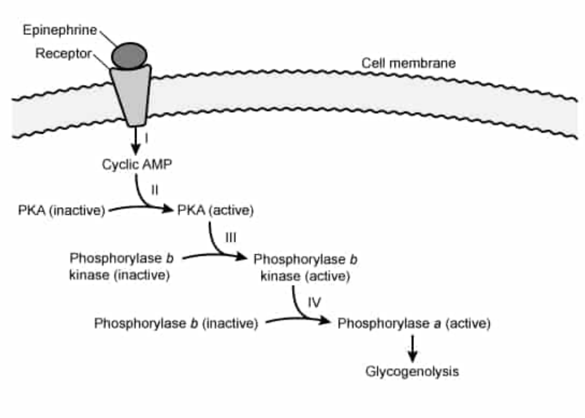


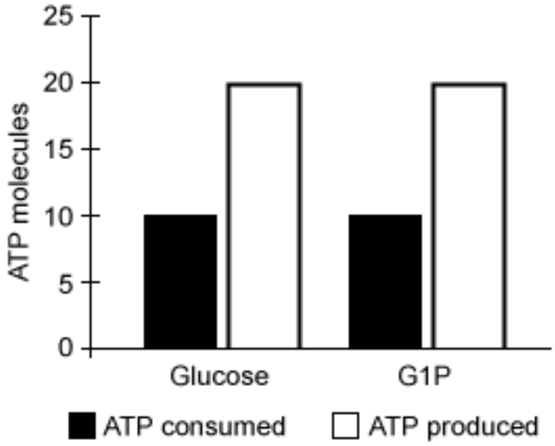
Figure 1 Stimulation of skeletal muscle glycogenolysis by epinephrine.

The relative signal amplification (SA) at each step (I–IV in Figure 1) can be determined by comparing the number of active signaling molecules before (M_{before}) and after (M_{after}) the step:

$$\text{SA} = \frac{M_{\text{after}}}{M_{\text{before}}}$$

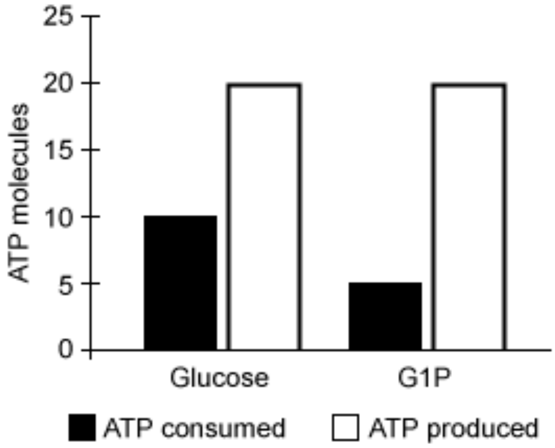
Each of the individual residues released during glycogenolysis can eventually enter glycolysis. Residues that are released by glycogen phosphorylase enter glycolysis after phosphoglucomutase converts them from glucose 1-phosphate (G1P) to glucose 6-phosphate (G6P). The remaining residues enter glycolysis when hexokinase converts them from glucose to glucose 6-phosphate. Which figure shows the correct comparison of ATP consumption and production when five residues enter glycolysis starting as glucose versus those that start as G1P?

○ A.


25
20
15
10
5
0
ATP molecules
Glucose
G1P
ATP consumed
ATP produced

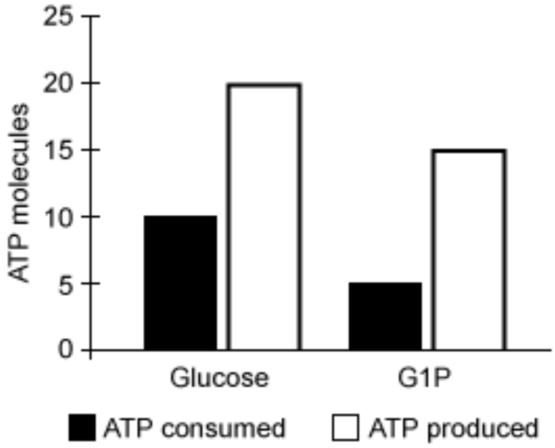
[10%]

B.


25
20
15
10
5
0
ATP molecules
Glucose
G1P
ATP consumed
ATP produced

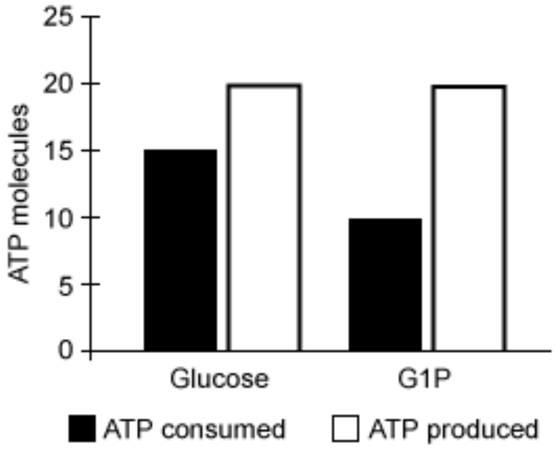
[65%]

C.


25
20
15
10
5
0
ATP molecules
Glucose
G1P
ATP consumed
ATP produced

[15%]

D.

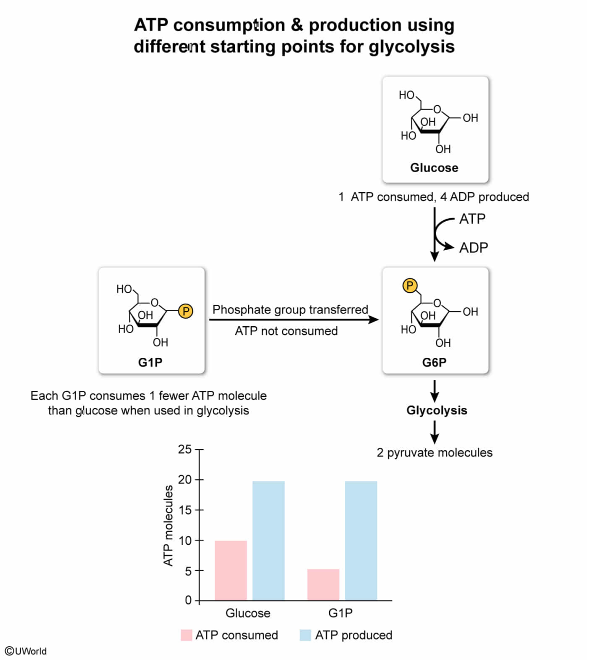

25
20
15
10
5
0
ATP molecules
Glucose
G1P
ATP consumed
ATP produced

[9%]

Correct

64% answered correctly

Explanation:



Glycolysis consists of the ten reactions by which glucose is converted to pyruvate. The reactions are divided into two phases: the energy investment phase, in which ATP is consumed, and the energy payoff phase, in which ATP is produced.

The passage describes the process of glycogenolysis in skeletal muscle, and the question notes that all the glucosyl residues released during glycogenolysis can enter glycolysis. The question asks for the figure showing the expected ATP use and release for 5 glycogen residues entering glycolysis as glucose compared to 5 molecules that start as glucose 1-phosphate (G1P).

Glycosyl residues entering glycolysis **as glucose** go through the **complete pathway**, with **2 ATP molecules invested** and **4 ATP molecules produced** per glucose. Glycosyl **residues entering** glycolysis **as G1P bypass** the first **hexokinase** reaction, and thus **require consumption of 1 less ATP** per residue during the investment phase.

Therefore, 5 glucose molecules consume a total of 10 ATP molecules (2 each) and produce a total of 20 (4 each), for a net gain of 10 ATP. In contrast, 5 G1P molecules consume a total of 5 ATP molecules (1 each) but still produce 20, for a net gain of 15 ATP. The figure in Choice B correctly depicts this outcome.

(Choice A) This figure incorrectly shows the consumption of 10 ATP molecules when G1P is the starting point.

(Choice C) The number of ATP molecules produced per G1P entering glycolysis is the same (ie, 4) as for glucose. Therefore, the total ATP produced from 5 molecules of G1P should be 20 (*not* 15).

(Choice D) The number of ATP molecules invested per glucose entering glycolysis is 2 (*not* 3), giving a total of 10 (*not* 15) for 5 glucose molecules entering glycolysis. Similarly, the number of ATP molecules invested per G1P entering glycolysis is 1 (*not* 2), giving a total of 5 (*not* 10) for 5 glucose molecules entering glycolysis.

Educational objective:
Glycolysis is the series of cytosolic reactions in which 1 glucose molecule is converted to 2 pyruvate molecules, normally resulting in the net production of 2 ATP molecules and 2 NADH molecules. When preformed glucose 6-phosphate enters glycolysis, less ATP is invested because the initial hexokinase reaction is bypassed.

Time spent:0 sec
Subject:
Biochemistry

QID:403535
System:
Metabolic Reactions

The body's largest stores of glycogen are found in the liver and skeletal muscles. Individual glucosyl residues (ie, glucose monomers) in muscle and liver glycogen particles are linked by α(1→4) and α(1→6) glycosidic bonds and are released in the reactions of glycogenolysis. Glycogen phosphorylase catalyzes the removal of most of the glucosyl residues, although debranching enzyme also releases the single glucose molecule that remains after it repositions three residues onto an adjacent linear chain.

Binding of the hormone epinephrine to its receptor on the surface of muscle cells stimulates muscle glycogenolysis through activation of a signaling cascade (Figure 1).

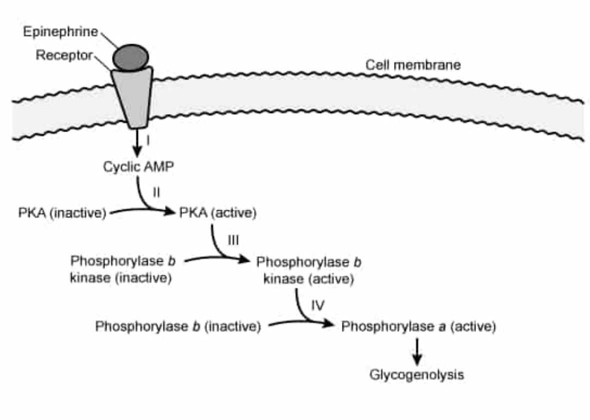


Figure 1 Stimulation of skeletal muscle glycogenolysis by epinephrine.

The relative signal amplification (SA) at each step (I–IV in Figure 1) can be determined by comparing the number of active signaling molecules before (M_{before}) and after (M_{after}) the step:

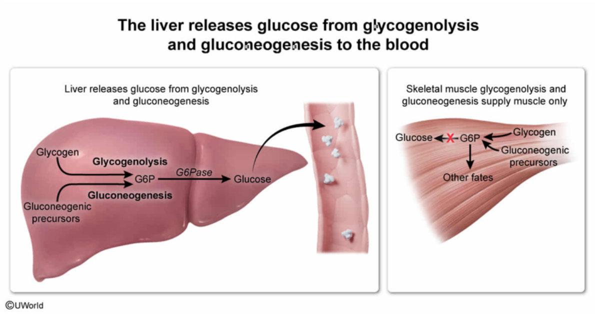
$$SA = \frac{M_{after}}{M_{before}}$$

The skeletal muscles collectively contain more than twice as much glucose stored in glycogen as is found in the liver. Compared with the liver, the amount of glucose released into the blood from muscles is likely:

- A. greater from glycogenolysis, but less from gluconeogenesis. [65%]
- B. the same from glycogenolysis, but more from gluconeogenesis. [8%]
- ✔ C. less from both glycogenolysis and gluconeogenesis. [23%]
- D. the same from both glycogenolysis and gluconeogenesis. [2%]

Correct
23% answered correctly

Explanation:



Blood glucose levels are determined by the handling of glucose stored or produced in the body. After dietary carbohydrates are digested and absorbed by the small intestine, they first pass through the liver, followed by entry into the systemic circulation for uptake by various tissues. After dietary carbohydrate absorption from the small intestine ceases (ie, during fasting or starvation), glucose derived from either stored glycogen or from gluconeogenic precursors enters the blood for uptake by tissues throughout the body.

The liver and the skeletal muscles contain most of the body's glycogen, and most glycogenolysis occurs in these two tissues. Much of the enzymatic machinery required for gluconeogenesis is also shared by liver and skeletal muscle. The question asks for the best characterization of the contributions of muscle glycogenolysis and gluconeogenesis to blood glucose levels, as compared with the two processes in the liver.

Because **skeletal muscle lacks** significant levels of **glucose 6-phosphatase**, it is **poorly equipped** to **convert glucose 6-phosphate (G6P)** derived from glycogenolysis or gluconeogenesis **into glucose** for **release into** the **blood**. Although a small proportion of the glucose residues released during skeletal muscle glycogenolysis are free (ie, unphosphorylated) glucose, these glucose molecules will likely be trapped inside the cell as G6P due to the activity of hexokinase. Therefore, **glucose release into** the **blood from muscle** glycogenolysis and gluconeogenesis (if any occurs) is **much less than from** the two processes in the **liver**.

(Choices A, B, and D) Skeletal muscle and liver play markedly different metabolic roles in the body, despite both being major reservoirs of glycogen and sharing many gluconeogenic reactions. Of the two tissues, only the liver expresses enough glucose 6-phosphatase to allow significant glucose release into the blood.

Educational objective:
Skeletal muscle and liver are the major sites of glycogen storage in the body. The activity of glucose 6-phosphatase allows the liver to release glucose derived from glycogenolysis and gluconeogenesis into the blood whereas muscle lacks significant levels of glucose 6-phosphatase.

Time spent:0 sec	**QID**:403536
Subject:	**System:**
Biochemistry	Metabolic Reactions